

Aaditya M. Kanase

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EDUCATION

North Carolina State University , Raleigh, NC	Current
• <i>Bachelor of Science in Electrical Engineering, Sophomore Standing</i>	
University of North Carolina at Charlotte , Charlotte, NC	May 2026
CGPA: 4.000/4.000, <i>Albert Engineering Leadership Scholar</i>	
Willow Spring High School , Fuquay-Varina, NC	June 2025
Graduated <i>Summa Cum Laude</i> , Class <i>Valedictorian</i>	

RESEARCH INTERESTS

- Physical Electronics
- Controls & Robotics
- Computational Analysis
- Intelligent Systems
- Signal Processing
- Energy storage & conversion

TECHNICAL SKILLS

Bioelectronics & Lab

- Nanofabrication, scRNA-seq, transcriptomics, Biomaterial handling.

Software & ML

- C++, Rust, Python (PyTorch, SciKit-Learn, OpenCV), ROS2, R-Studio

Hardware / Embedded

- VHDL, Arduino, ESP32, FPGA (Xilinx)
- FSM design, timing analysis, waveform interpretation, Analog circuit analysis
- SPICE, KiCad, Vivado

RESEARCH & PROJECTS

Control Optimization & Autonomy Robotics Lab August 2025 - Present

Mentored Student Researcher

- Constructed a dry-electrode EIT-based Robotic Teleoperation device
 - Designed a 16-electrode bioimpedance sensing system with signal processing and TinyML classification on ESP32; used for wireless robotic arm control via ESP-NOW.
- ROS-based Simulation of a *Partial Information Target Defense Game* [Dipankar Maity, IEEE 2021]
 - W.S.L College of Engineering Independent Study; Simulation implementation from scratch.

Microvascular Engineering and Regeneration Lab November 2025 - Present

Undergraduate Research Assistant

- Supported nanofabrication of microfluidic chip models that mimic microvascular systems for next-gen skin graft research.
- Large dataset single-cell sequencing (scSeq) identifying transcriptomic changes under differing angiogenic conditions to identify key features between iBMEC and iPC distributions in tissue-engineered brain post-capillary venules.

PUBLICATIONS

1. [Tracy D. Chung, Linus Wang, Alexander F. Pessel, Peter Searson, Nan Zhao, **Aaditya Kanase**], "Transcriptomic changes in a tissue-engineered brain post-capillary venule during angiogenesis," **Manuscript in preparation**, expected submission 2026. Contribution: single-cell RNA-seq (scRNA-seq) analysis; produced primary transcriptomic visualizations

RELEVANT COURSEWORK

Engineering : Electric Circuits; Computer Programming (C++); Logic Systems Design; General Physics II

Mathematics : Linear Algebra; Applied Differential Equations; Multivariable Calculus; Discrete Structures; Probability and Statistics